

Gestão de Sistemas e de Redes - L2S1H

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1.

O ZFS é um sistema de arquivos desenvolvido pela Sun Microsystems (atual Oracle) para uso no sistema operativo Unix desenvolvido com o propósito de solucionar uma série de problemas existentes em outros sistemas de arquivos. Considerado atualmente uma solução revolucionária de armazenamento, o ZFS apresenta facilidade de gerenciamento e robustez.

As vantagens do ZFS são:

- Facilidade no acesso de dados;
- Verificação automática de pedidos;
- Gerenciamento de RAID por software e LVM no sistema de arquivos, o COW (Copy-On-Write) torna as operações de escrita de dados em locais físicos mais segura e potente e determinação de prioridades nas operações entrada/saída.

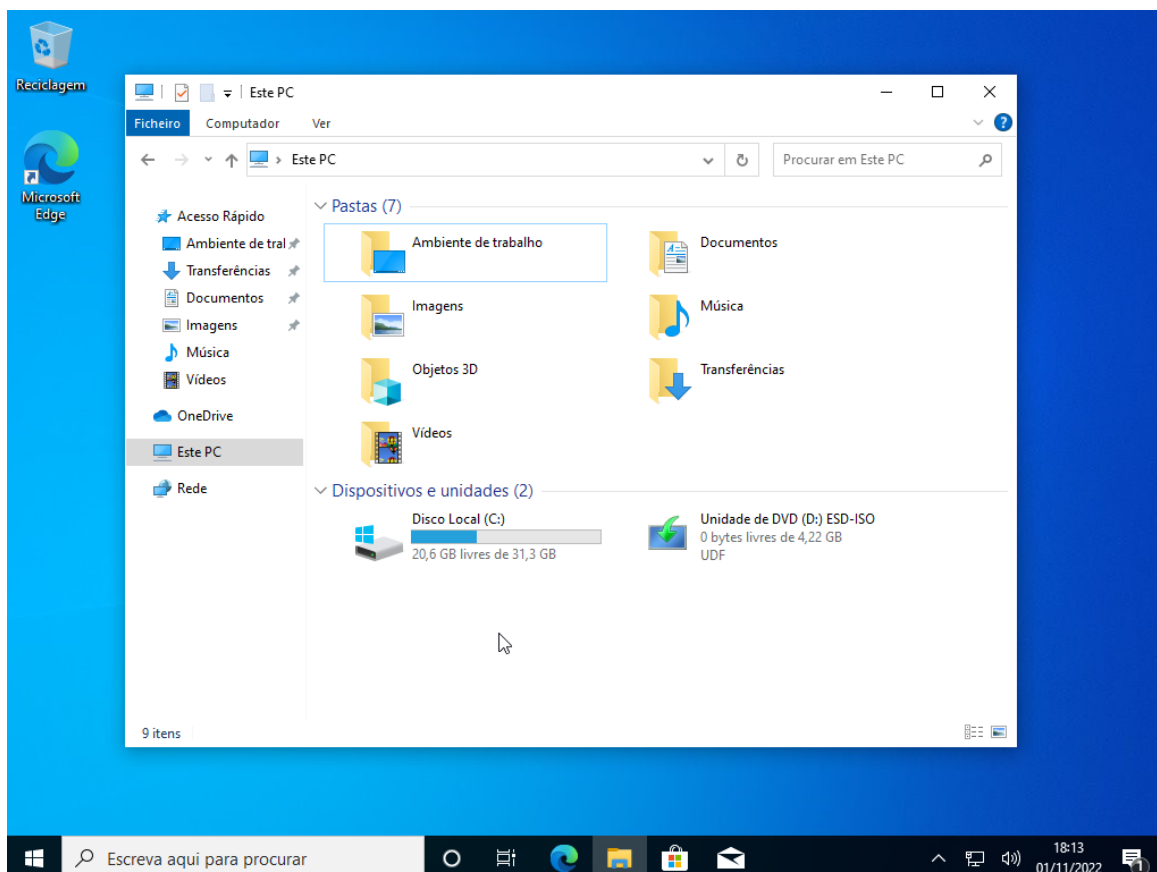
Uma desvantagem deste sistema é ser limitado a algumas plataformas. Por ter sido criado originalmente para o Unix (Solaris), o sistema era compatível apenas para empresas com servidores poderosos, neste momento é já compatível com outros sistemas operativos como Linux, Illumos e BSD.

2.

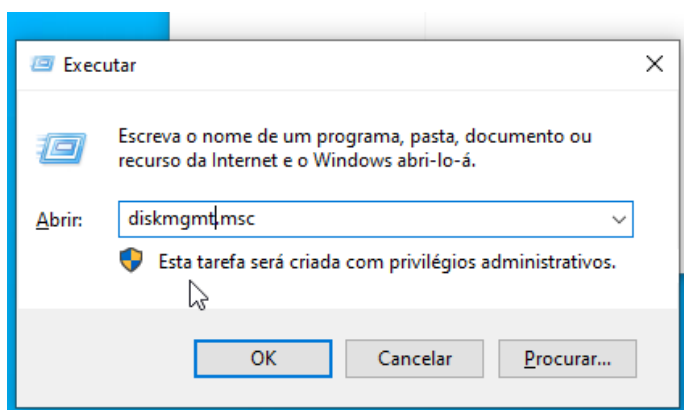
Hot spare é uma unidade que atua como unidade de reserva no volume RAID 1, RAID 5 ou RAID 6. É uma unidade totalmente funcional que não contém dados e não é usada durante a operação normal. Se uma unidade do volume falhar, o controlador reconstrói os dados da unidade com falha para a unidade de hot spare.

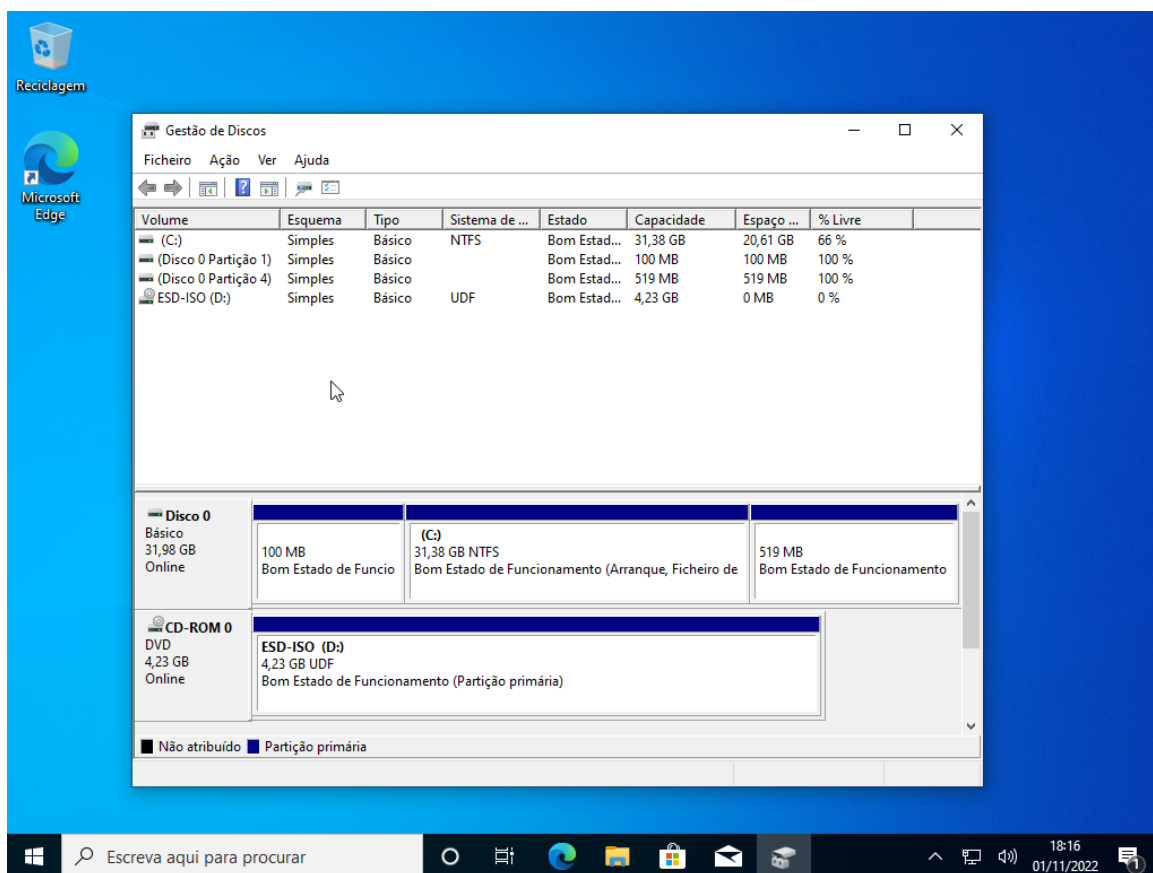
3.

- Instalar windows 10 Pro em modo administrador, saltar o Out-of the box experience, para uma instalação mais limpa.

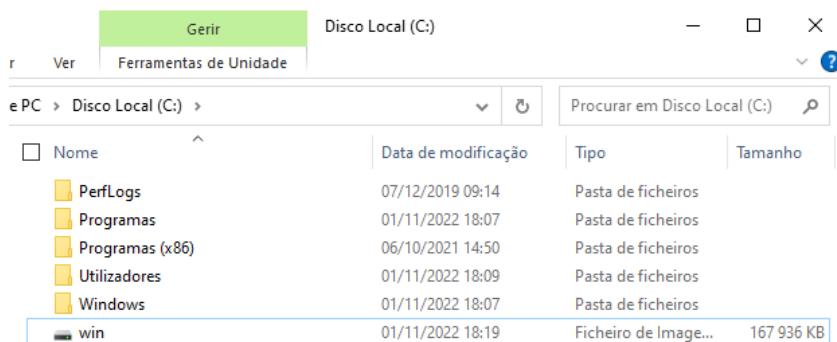
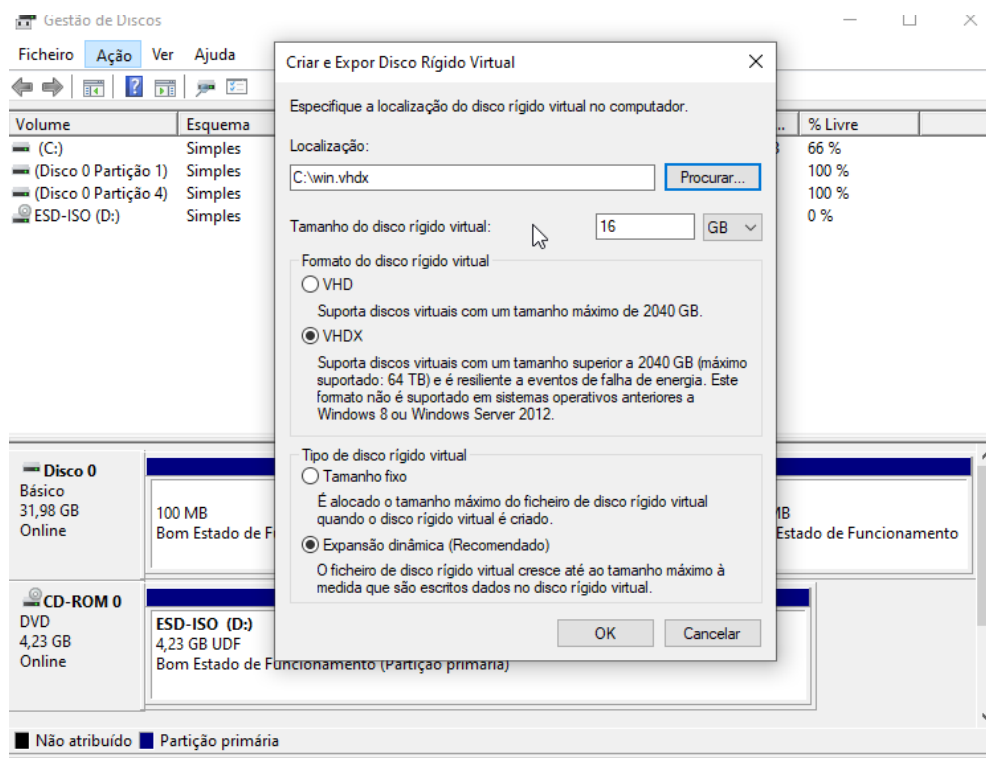


- Executar gestor de discos

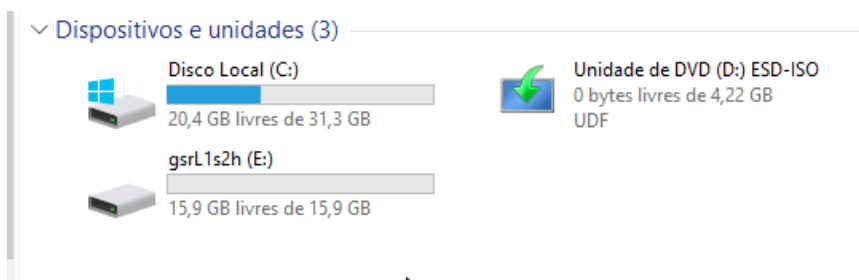
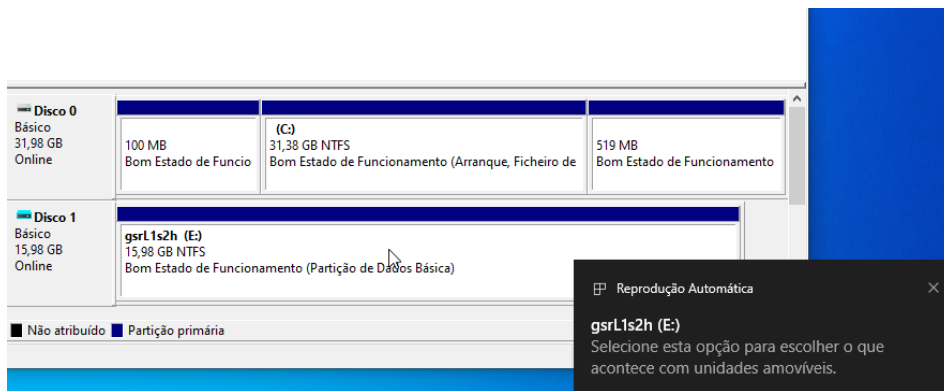




- Criar um novo Virtual hard drive (vhdx de 16gb) no disco Local C:\



-Inicializar o disco e criar um novo volume simples



- Converter install.esd para install.wim, a partir da pasta sources da unidade de instalação do windows (D:\)

Descobrir quais imagens do windows 10 estão no arquivo install.esd;

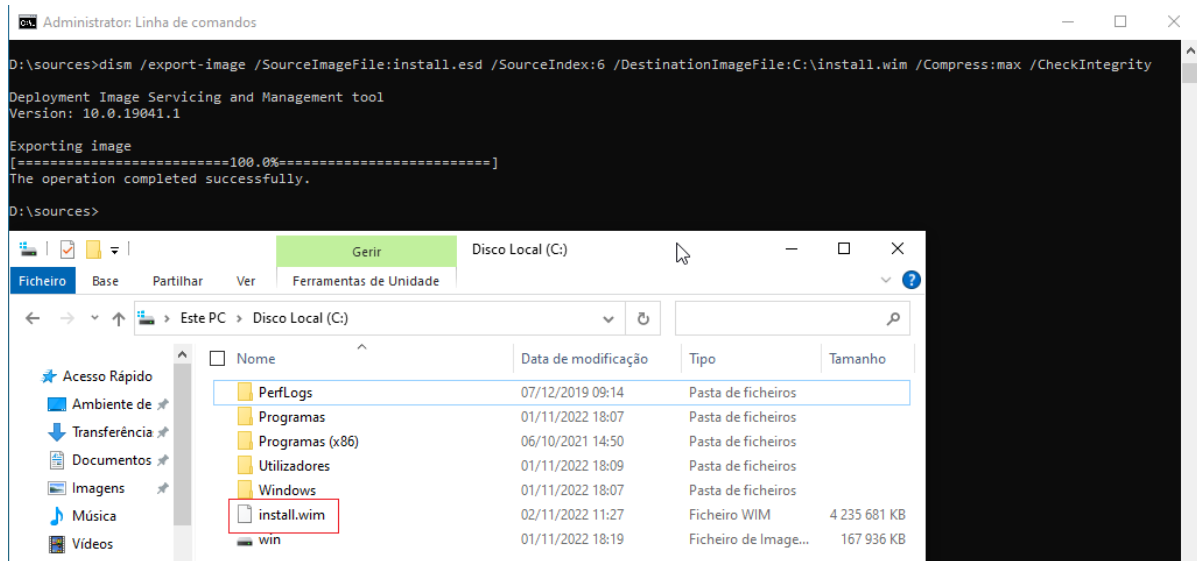
```
D:\sources> dism /Get-WimInfo /WimFile:install.esd
```

Selecionar o índice que contém a mesma versão do windows 10 instalada.

```
Index : 6
Name : Windows 10 Pro
Description : Windows 10 Pro
Size : 14 875 975 652 bytes
```

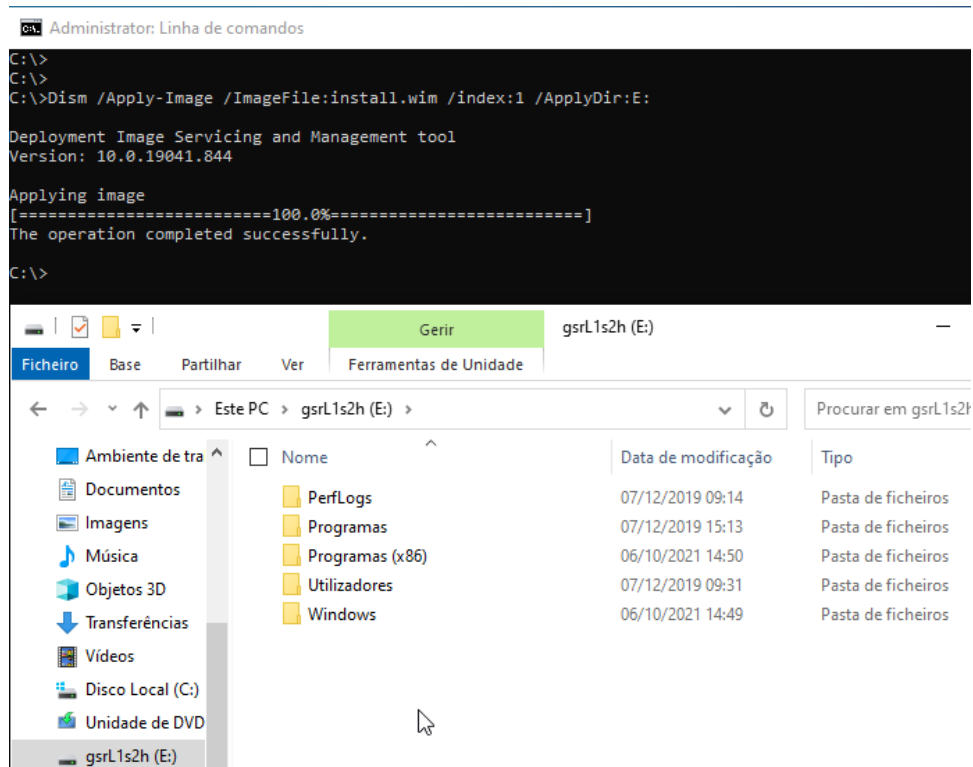
- Extrair o arquivo install.wim a partir do install.esd correspondente com a versão do windows (index 6) e guardar no disco local (C:\)

```
D:\sources> dism /export-image /SourceImageFile:install.esd  
/SourceIndex:6 /DestinationImageFile:C:\install.wim \Compress:max  
\CheckIntegrity
```

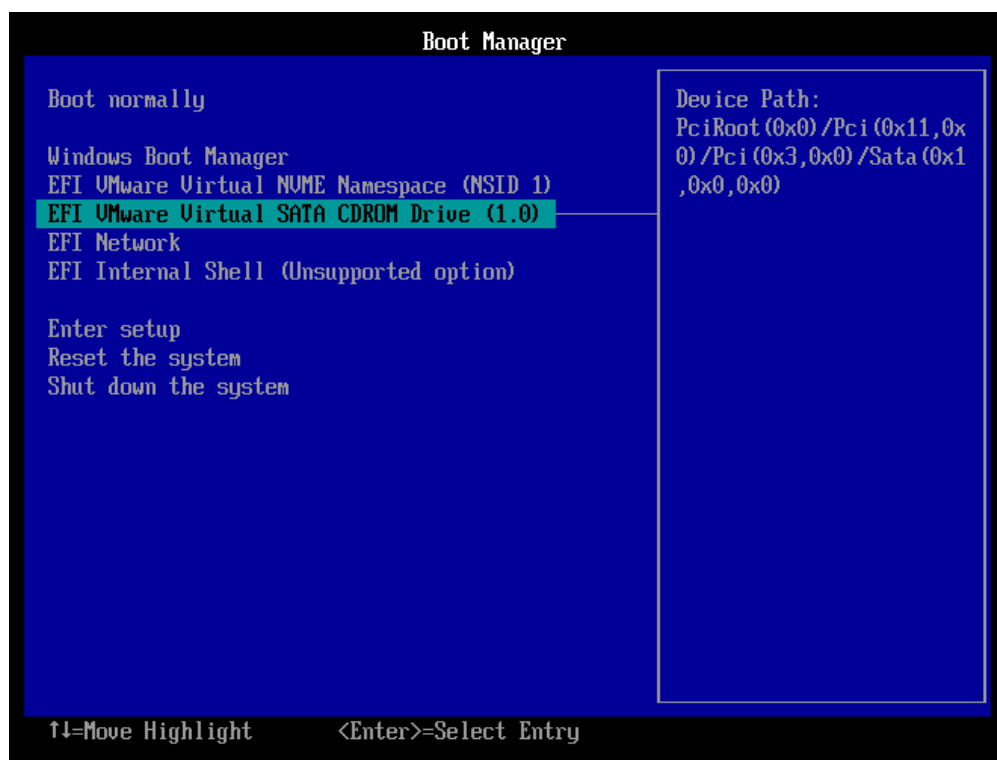


-Aplicar o arquivo install.wim para o vhd (E:\).

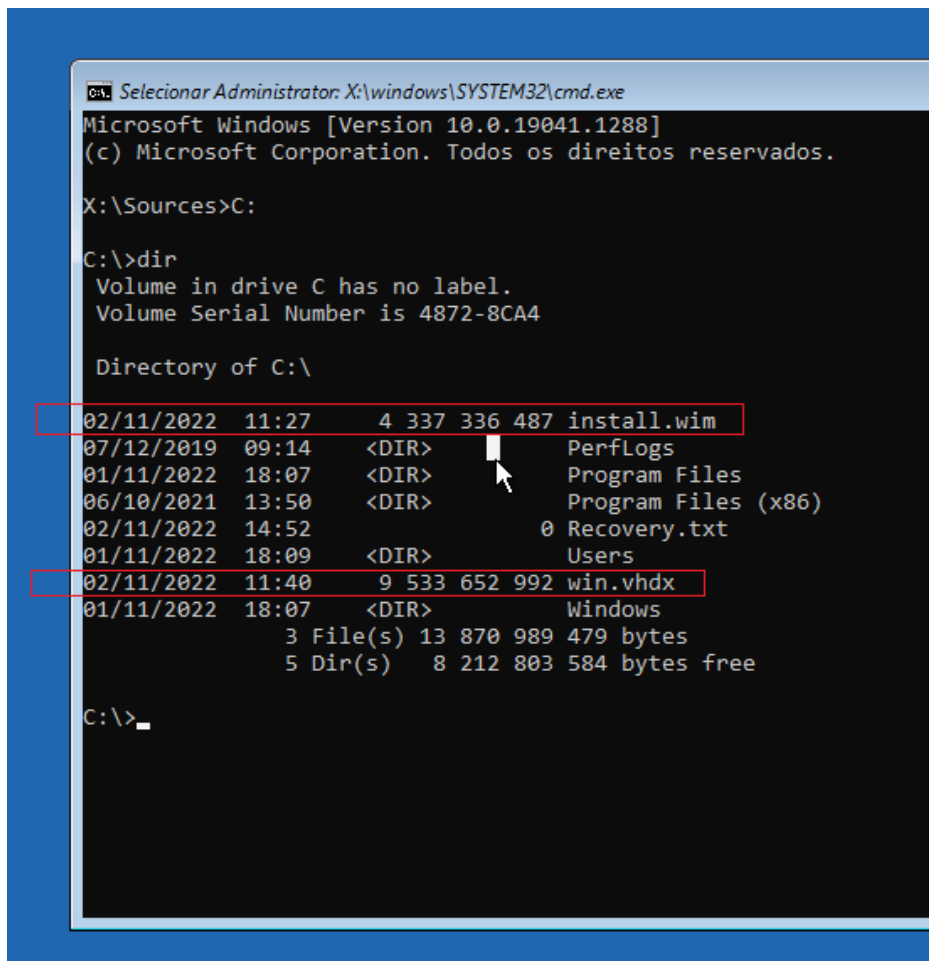
```
C:\> dism /Apply-Image /ImageFile:install.wim /index:1 /ApplyDir:E:
```



-Reiniciar o windows, fazer o boot e seleccionar o drive que contem a instalação do windows



-Selecionar reparar computador em vez de instalar e abrir a consola:



The screenshot shows a Windows command prompt window titled "Selecionar Administrador: X:\windows\SYSTEM32\cmd.exe". The prompt is at "X:\Sources>C:". The user has entered the command "C:\>dir". The output shows the directory contents of drive C:.

```
Microsoft Windows [Version 10.0.19041.1288]
(c) Microsoft Corporation. Todos os direitos reservados.

X:\Sources>C:

C:\>dir
Volume in drive C has no label.
Volume Serial Number is 4872-8CA4

Directory of C:\

02/11/2022  11:27    4 337 336 487 install.wim
07/12/2019  09:14    <DIR>          PerfLogs
01/11/2022  18:07    <DIR>          Program Files
06/10/2021  13:50    <DIR>          Program Files (x86)
02/11/2022  14:52             0 Recovery.txt
01/11/2022  18:09    <DIR>          Users
02/11/2022  11:40    9 533 652 992 win.vhdx
01/11/2022  18:07    <DIR>          Windows
               3 File(s) 13 870 989 479 bytes
               5 Dir(s)  8 212 803 584 bytes free

C:\>_
```

- Usar o comando diskpart que nos permite editar partições, volumes e os próprios discos.
- Selecionar o disco virtual vhdx criado. Anexar esse disco que ele apareça no computador host como uma unidade de disco rígido local.
- Selecionar o volume 2.


```
Administrator: X:\windows\SYSTEM32\cmd.exe - diskpart
C:\>diskpart

Microsoft DiskPart version 10.0.19041.964

Copyright (C) Microsoft Corporation.
On computer: MININT-J351TRE

DISKPART> select vdisk file=C:\win.vhdx
DiskPart successfully selected the virtual disk file.

DISKPART> attach vdisk

100 percent completed

DiskPart successfully attached the virtual disk file.

DISKPART> list volume

Volume ### Ltr Label Fs Type Size Status Info
-----
Volume 0 D ESD-ISO UDF DVD-ROM 4331 MB Healthy
Volume 1 C NTFS Partition 31 GB Healthy
Volume 2 FAT32 Partition 100 MB Healthy Hidden
Volume 3 NTFS Partition 519 MB Healthy Hidden
Volume 4 E gsrL1s2h NTFS Partition 15 GB Healthy

DISKPART> select volume 2
Volume 2 is the selected volume.

DISKPART> _
```

-Atribuir a letra Z ao volume

```
DISKPART> select volume 2
Volume 2 is the selected volume.

DISKPART> assign letter=Z
DiskPart successfully assigned the drive letter or mount point.

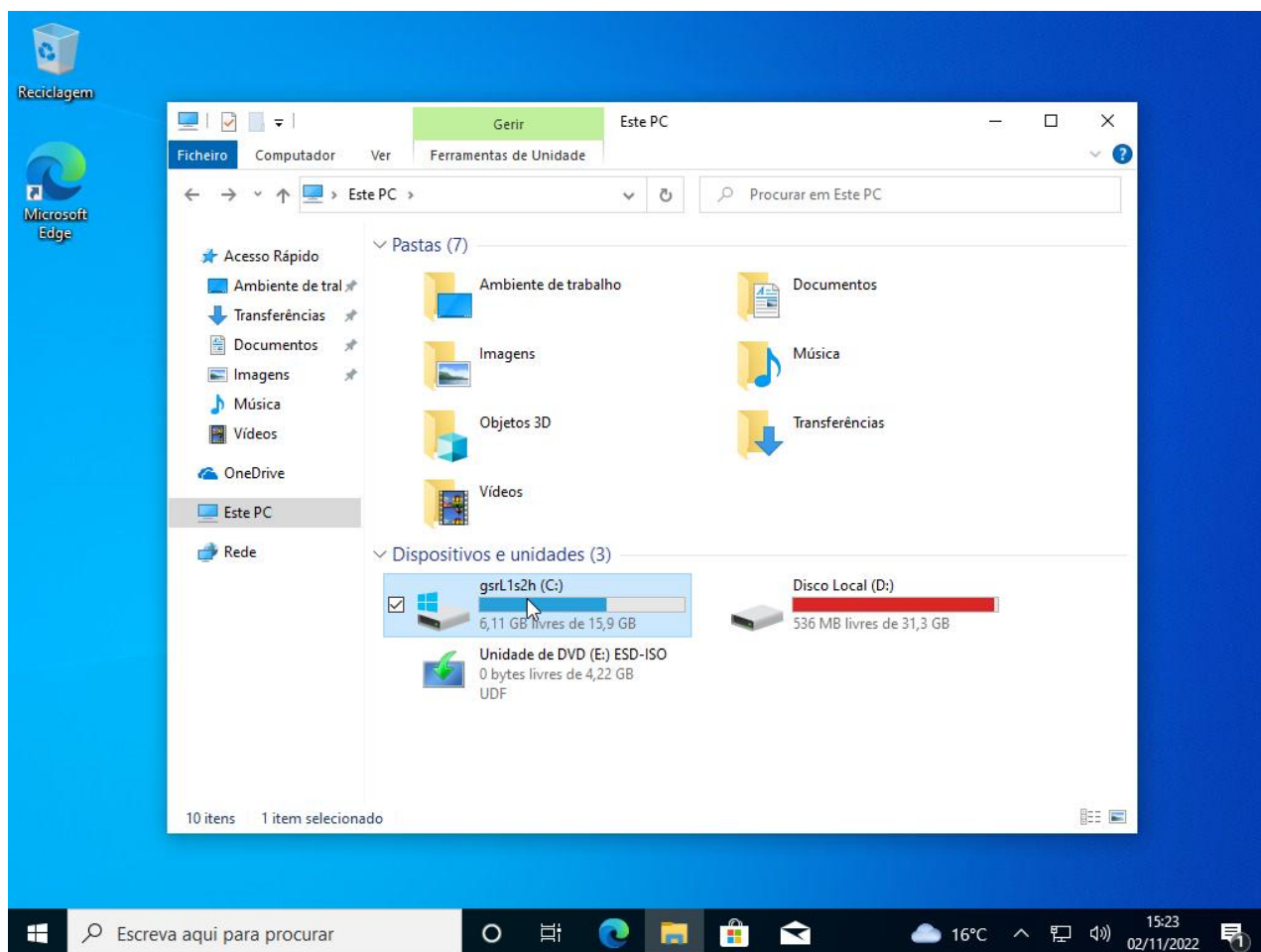
DISKPART> _
```

- configurar os arquivos de inicialização executar o sistema operacional Windows.

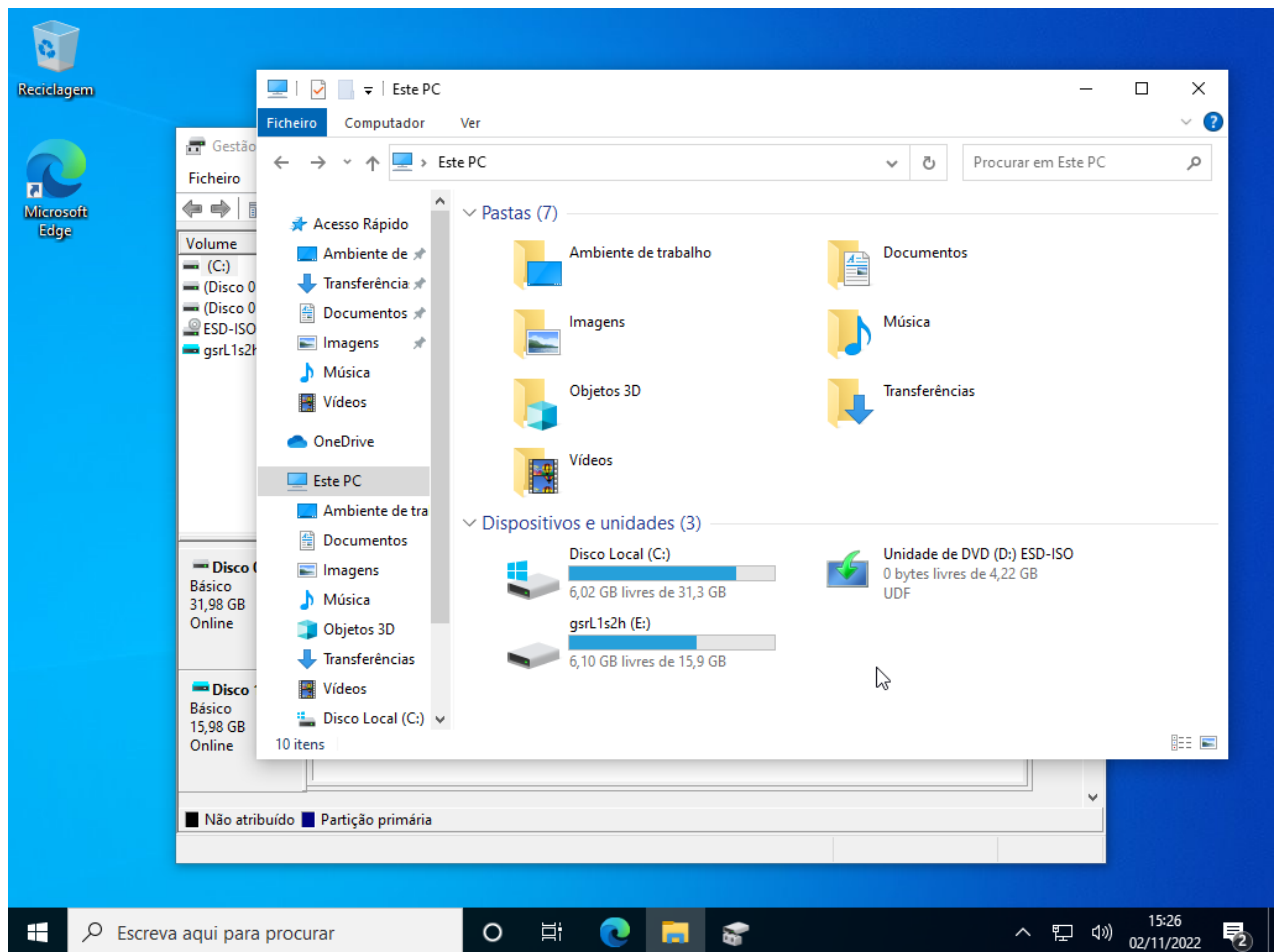
```
C:\>bcdboot E:\windows /s Z: /f UEFI
Boot files successfully created.

C:\>_
```

-Sair da console continuar no volume selecionado e proceder à instalação do Windows no volume;

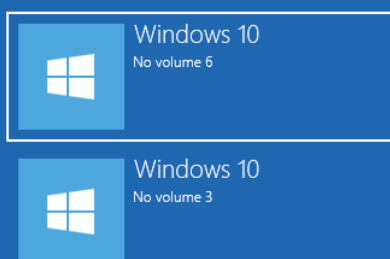


- Outro windows:



Ao iniciar o windows, possibilidade do dual boot.

Escolher um sistema operativo



Alterar as predefinições ou escolher outras opções

4.

a)

```
rafaelswr@gsr:~$ sudo mdadm -V
mdadm - v4.1 - 2018-10-01
rafaelswr@gsr:~$ lsblk -o NAME,SIZE
NAME        SIZE
sda          102M
sdb          45G
├─sdb1       8,7G
├─sdb2        1K
├─sdb5       3,2G
├─sdb6       976M
├─sdb7       632M
└─sdb8      31,5G
sdc          102M
sdd          102M
sde          102M
sdf          102M
sdg          102M
rafaelswr@gsr:~$ █
```

```

rafaelswr@gsr:~$ sudo mdadm --create --verbose /dev/md0 --level=6 --raid-devices
=4 /dev/sda /dev/sdc /dev/sdd /dev/sde --spare-devices=2 /dev/sdf /dev/sdg
mdadm: layout defaults to left-symmetric
mdadm: layout defaults to left-symmetric
mdadm: chunk size defaults to 512K
mdadm: size set to 102400K
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
rafaelswr@gsr:~$ █

```

b)

```

rafaelswr@gsr:~$ sudo mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov  8 20:23:32 2022
    Raid Level : raid6
    Array Size : 204800 (200.00 MiB 209.72 MB)
  Used Dev Size : 102400 (100.00 MiB 104.86 MB)
    Raid Devices : 4
  Total Devices : 6
 Persistence : Superblock is persistent

    Update Time : Tue Nov  8 20:23:33 2022
      State : clean
 Active Devices : 4
Working Devices : 6
Failed Devices : 0
Spare Devices : 2


    Layout : left-symmetric
    Chunk Size : 512K

Consistency Policy : resync

        Name : gsr:0 (local to host gsr)
        UUID : 7f14b693:b71fela2:6764761f:4f771cec
        Events : 17

   Number  Major   Minor  RaidDevice State
     0       8       0           0  active sync  /dev/sda
     1       8      32           1  active sync  /dev/sdc
     2       8      48           2  active sync  /dev/sdd
     3       8      64           3  active sync  /dev/sde
     4       8      80           -    spare      /dev/sdf
     5       8      96           -    spare      /dev/sdg
rafaelswr@gsr:~$ █

```

c)

Device	Summary
 Memory	2.1 GB
 Processors	2
 Hard Disk (SCSI)	45 GB
 Hard Disk 4 (SCSI)	102 MB
 Hard Disk 5 (SCSI)	102 MB
 Hard Disk 7 (SCSI)	102 MB
 Hard Disk 6 (SCSI)	102 MB
 Hard Disk 2 (SCSI)	102 MB
 Hard Disk 3 (SCSI)	102 MB
 New Hard Disk (SCSI)	204 MB
 New Hard Disk (SCSI)	204 MB
 Network Adapter	NAT
 USB Controller	Present
 Sound Card	Auto detect
 Display	Auto detect

d)

```
$ sudo apt install lvm2

rafaelswr@gsr:~$ sudo pvcreate /dev/md0
Physical volume "/dev/md0" successfully created.
rafaelswr@gsr:~$ sudo pvs
PV          VG Fmt  Attr PSize  PFree
/dev/md0    lvm2 --- 200,00m 200,00m
rafaelswr@gsr:~$ sudo pvcreate /dev/sdh
Physical volume "/dev/sdh" successfully created.
rafaelswr@gsr:~$ sudo pvcreate /dev/sdi
Physical volume "/dev/sdi" successfully created.
rafaelswr@gsr:~$ █

rafaelswr@gsr:~$ sudo pvs
PV          VG Fmt  Attr PSize  PFree
/dev/md0    lvm2 --- 200,00m 200,00m
/dev/sdh    lvm2 --- 204,00m 204,00m
/dev/sdi    lvm2 --- 204,00m 204,00m
rafaelswr@gsr:~$ █
```

e)

```
rafaelswr@gsr:~$ sudo vgcreate vg0 /dev/md0 /dev/sdh
Volume group "vg0" successfully created
rafaelswr@gsr:~$ sudo vgs
VG   #PV #LV #SN Attr   VSize   VFree
vg0   2   0   0 wz--n- 396,00m 396,00m
rafaelswr@gsr:~$
```

f)

```
rafaelswr@gsr:~$ sudo pvs
PV          VG   Fmt  Attr  PSize   PFree
/dev/md0    vg0  lvm2 a--   196,00m 196,00m
/dev/sdh    vg0  lvm2 a--   200,00m 200,00m
/dev/sdi          lvm2 ---   204,00m 204,00m
rafaelswr@gsr:~$ sudo lvcreate -L 200M vg0 -n lv0
Logical volume "lv0" created.
rafaelswr@gsr:~$ sudo lvs
LV   VG   Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
lv0  vg0  -wi-a----- 200,00m
rafaelswr@gsr:~$ sudo vgdisplay vg0
--- Volume group ---
VG Name                vg0
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   2
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 1
Open LV                 0
Max PV                 0
Cur PV                 2
Act PV                 2
VG Size                 396,00 MiB
PE Size                 4,00 MiB
Total PE                99
Alloc PE / Size         50 / 200,00 MiB
Free PE / Size           49 / 196,00 MiB
VG UUID                 UNS0Y6-9i4h-mUyW-Ra06-oQaW-FQm1-vViu21
```

g)

```
$ sudo apt install jfsutils
```

-> Criar o diretorio /mnt/vg0-lv0 e instalar o *Journaled File System* no lv0.

```
rafaelswr@gsr:~$ sudo mkdir /mnt/vg0-lv0
rafaelswr@gsr:~$ sudo mkfs.jfs /dev/vg0/lv0
mkfs.jfs version 1.1.15, 04-Mar-2011
Warning! All data on device /dev/vg0/lv0 will be lost!
```

Continue? (Y/N) Y

Format completed successfully.

204800 kilobytes total disk space.

```
rafaelswr@gsr:~$ █
```

-> comando blkid para exibir todas as partições do disco e seu UUID associado

```
rafaelswr@gsr:~$ sudo blkid
/dev/sdd: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="569d700d-b9ab-a1fe-89cd-8ff861087b0b" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sdc: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="cb0217a2-21a0-38e0-ac16-ca68f62428ff" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sde: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="66917d97-1301-4433-75ae-c08e22375c32" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sdb1: UUID="03cb6f4d-865b-417a-a6ff-c332b3ed194d" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="0e0daff0-01"
/dev/sdb5: UUID="f6fd9fa6-9622-4b1b-af40-ed86bfc12a0b" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="0e0daff0-05"
/dev/sdb6: UUID="f658f933-a907-4ed3-8067-745a46805a0c" TYPE="swap" PARTUUID="0e0daff0-06"
/dev/sdb7: UUID="bd653c81-8f2a-4f34-8c1e-20bd23d46340" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="0e0daff0-07"
/dev/sdb8: UUID="f42af4fa-467e-4cef-9e96-8d0d19e75384" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="0e0daff0-08"
/dev/sdg: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="0be97a80-0695-205a-172e-b4de4c843b51" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sda: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="6091a84f-845a-d193-a9c9-d1b0fcc04362" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sdf: UUID="97b7359b-e924-79ea-63e6-628e54f91545" UUID_SUB="600576f1-1617-9048-2e0f-70a6f8e711a3" LABEL="gsr:0" TYPE="linux_raid_member"
/dev/sdi: UUID="Ytjppj0-xbRU-r4v3-nX6Y-rNa0-a6WK-C7DMqq" TYPE="LVM2_member"
/dev/sdh: UUID="zXezkf-Mie2-SUaX-pNjW-8CYK-hcVG-L7fXom" TYPE="LVM2_member"
/dev/md0: UUID="UuGC58-nSLu-7ATn-0l2n-CnVu-wzPf-pM0wzJ" TYPE="LVM2_member"
/dev/mapper/vg0-lv0: UUID="a34aa8b6-529d-4b80-a127-c1824023e8ab" BLOCK_SIZE="4096" TYPE="jfs"
rafaelswr@gsr:~$ █
```

-> O arquivo fstab no diretório /etc/ controla como os sistemas de arquivos são montados durante as operações de inicialização do sistema operacional. Ele contém uma lista de partições, pontos de montagem e opções de montagem.

Editar e guardar no arquivo /etc/fstab uma nova linha para a nova instrução de montagem automática.


```

GNU nano 5.4 /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=03cb6f4d-865b-417a-a6ff-c332b3ed194d / ext4 errors=remount-ro 0
# /home was on /dev/sda8 during installation
UUID=f42af4fa-467e-4cef-9e96-8d0d19e75384 /home ext4 defaults 0
# /tmp was on /dev/sda7 during installation
UUID=bd653c81-8f2a-4f34-8c1e-20bd23d46340 /tmp ext4 defaults 0
# /var was on /dev/sda5 during installation
UUID=f6fd9fa6-9622-4b1b-af40-ed86bfc12a0b /var ext4 defaults 0
# swap was on /dev/sda6 during installation
UUID=f658f933-a907-4ed3-8067-745a46805a0c none swap sw 0
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
UUID="a34aa8b6-529d-4b80-a127-c1824023e8ab" /mnt/vg0-lv0 jfs defaults 0 0

[ 23 linhas escritas ]
^G Ajuda ^O Gravar ^W Procurar ^K Cortar ^T Executar ^C Localização
^X Sair ^R Carregar ^_ Substituir ^U Colar ^J Justificar ^I Ir p/ linha

```

-> Usar o comando *mount* com a opção -a para montar todas as entradas no arquivo /etc/fstab

```

rafaelswr@gsr:~$ sudo mount -a
rafaelswr@gsr:~$

```

-> Usar o utilitário *df* para verificar o sistema de ficheiros montado:

```

rafaelswr@gsr:~$ sudo df -h
Sistema de ficheiros Tamanho      Uso Livre  Uso% Montado em
udev                999M          0  999M   0% /dev
tmpfs               205M        1,5M  204M   1% /run
/dev/sdb1           8,5G        4,0G   4,1G  50% /
tmpfs              1023M          0 1023M   0% /dev/shm
tmpfs               5,0M        4,0K   5,0M   1% /run/lock
/dev/sdb7           606M       116K   561M   1% /tmp
/dev/sdb8           31G         21M    30G   1% /home
/dev/sdb5           3,1G       464M   2,5G  16% /var
tmpfs               205M       136K   205M   1% /run/user/1000
/dev/mapper/vg0-lv0 199M       160K   199M   1% /mnt/vg0-lv0
rafaelswr@gsr:~$

```

h)

```
rafaelswr@gsr:~$ sudo vgdisplay vg0
--- Volume group ---
VG Name                vg0
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   2
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                1
Open LV                1
Max PV                 0
Cur PV                2
Act PV                 2
VG Size                396,00 MiB
PE Size                4,00 MiB
Total PE               99
Alloc PE / Size        50 / 200,00 MiB
Free PE / Size         49 / 196,00 MiB
VG UUID                aHEuHI-0los-H0pw-GScG-wQUf-sERz-T3odA5

rafaelswr@gsr:~$ sudo vgextend vg0 /dev/sdi
Volume group "vg0" successfully extended
rafaelswr@gsr:~$ █

rafaelswr@gsr:~$ sudo vgdisplay vg0
--- Volume group ---
VG Name                vg0
System ID
Format                 lvm2
Metadata Areas         3
Metadata Sequence No   3
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                1
Open LV                1
Max PV                 0
Cur PV                3
Act PV                 3
VG Size                596,00 MiB
PE Size                4,00 MiB
Total PE               149
Alloc PE / Size        50 / 200,00 MiB
Free PE / Size         99 / 396,00 MiB
VG UUID                aHEuHI-0los-H0pw-GScG-wQUf-sERz-T3odA5

rafaelswr@gsr:~$ █
```

i)

```
rafaelswr@gsr:~$ sudo vgdisplay
--- Volume group ---
VG Name                vg0
System ID
Format                 lvm2
Metadata Areas         3
Metadata Sequence No   3
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                1
Open LV                1
Max PV                 0
Cur PV                3
Act PV                 3
VG Size                596,00 MiB
PE Size                4,00 MiB
Total PE               149
Alloc PE / Size        50 / 200,00 MiB
Free PE / Size         99 / 396,00 MiB
VG UUID                aHEuHI-0los-H0pw-GScG-wQUf-sERz-T3odA5
```

```
rafaelswr@gsr:~$ sudo lvcreate -L 52M -m1 -n lv1 vg0
Logical volume "lv1" created.
```

```
rafaelswr@gsr:~$ █
```

```
rafaelswr@gsr:~$ sudo vgdisplay
--- Volume group ---
VG Name                vg0
System ID
Format                 lvm2
Metadata Areas         3
Metadata Sequence No   5
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                2
Open LV                1
Max PV                 0
Cur PV                3
Act PV                 3
VG Size                596,00 MiB
PE Size                4,00 MiB
Total PE               149
Alloc PE / Size        78 / 312,00 MiB
Free PE / Size         71 / 284,00 MiB
VG UUID                aHEuHI-0los-H0pw-GScG-wQUf-sERz-T3odA5
```

```
rafaelswr@gsr:~$ █
```

j)

```
rafaelswr@gsr:~$ sudo lvcreate -i2 -I64 -L190M -n lv2 vg0
Rounding up size to full physical extent 192,00 MiB
Logical volume "lv2" created.
```

```
rafaelswr@gsr:~$ █
```

k)

```

rafaelswr@gsr:~$ sudo lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Conve
rt
lv0 vg0 -wi-ao---- 200,00m
lv1 vg0 rwi-a-r--- 52,00m 100,00
lv2 vg0 -wi-a----- 192,00m

rafaelswr@gsr:~$ sudo df -h
Sistema de ficheiros Tamanho Uso Livre Uso% Montado em
udev 999M 0 999M 0% /dev
tmpfs 205M 1,5M 204M 1% /run
/dev/sdb1 8,5G 4,0G 4,1G 50% /
tmpfs 1023M 0 1023M 0% /dev/shm
tmpfs 5,0M 4,0K 5,0M 1% /run/lock
/dev/sdb7 606M 116K 561M 1% /tmp
/dev/sdb8 31G 21M 30G 1% /home
/dev/sdb5 3,1G 464M 2,5G 16% /var
tmpfs 205M 224K 205M 1% /run/user/1000
/dev/mapper/vg0-lv0 199M 160K 199M 1% /mnt/vg0-lv0
rafaelswr@gsr:~$ sudo lvextend -L+60M /dev/vg0/lv0
Size of logical volume vg0/lv0 changed from 200,00 MiB (50 extents) to 260,00
MiB (65 extents).
Logical volume vg0/lv0 successfully resized.
rafaelswr@gsr:~$

rafaelswr@gsr:~$ sudo mount -o remount,resize /mnt/vg0-lv0
rafaelswr@gsr:~$ sudo lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Conve
rt
lv0 vg0 -wi-ao---- 260,00m
lv1 vg0 rwi-a-r--- 52,00m 100,00
lv2 vg0 -wi-a----- 192,00m

rafaelswr@gsr:~$ sudo df -h
Sistema de ficheiros Tamanho Uso Livre Uso% Montado em
udev 999M 0 999M 0% /dev
tmpfs 205M 1,5M 204M 1% /run
/dev/sdb1 8,5G 4,0G 4,1G 50% /
tmpfs 1023M 0 1023M 0% /dev/shm
tmpfs 5,0M 4,0K 5,0M 1% /run/lock
/dev/sdb7 606M 116K 561M 1% /tmp
/dev/sdb8 31G 21M 30G 1% /home
/dev/sdb5 3,1G 464M 2,5G 16% /var
tmpfs 205M 224K 205M 1% /run/user/1000
/dev/mapper/vg0-lv0 258M 168K 258M 1% /mnt/vg0-lv0
rafaelswr@gsr:~$

```

l)

->Visão geral do array md0, com /sdf e /sdg marcados como Spare.

->Causando a falha no /sda, é marcado como F, e o sdg pata para o estado ativo, ficando o /sdf como Spare.

```

rafaelswr@gsr:~$ sudo cat /proc/mdstat
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1] [r
aid10]
md0 : active raid6 sdg[5](S) sdf[4](S) sde[3] sdd[2] sdc[1] sda[0]
      204800 blocks super 1.2 level 6, 512k chunk, algorithm 2 [4/4] [UUUU]

unused devices: <none>
rafaelswr@gsr:~$ sudo mdadm --manage /dev/md0 --fail /dev/sda
mdadm: set /dev/sda faulty in /dev/md0
rafaelswr@gsr:~$ sudo cat /proc/mdstat
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1] [r
aid10]
md0 : active raid6 sdg[5] sdf[4](S) sde[3] sdd[2] sdc[1] sda[0](F)
      204800 blocks super 1.2 level 6, 512k chunk, algorithm 2 [4/4] [UUUU]

unused devices: <none>
rafaelswr@gsr:~$ █

```

-> remover o /sda do array, ficando o /sdc, /sdd, /sde e /sdg como ativos e apenas o /sdf como Spare.

```

rafaelswr@gsr:~$ sudo mdadm --manage /dev/md0 --remove /dev/sda
mdadm: hot removed /dev/sda from /dev/md0
rafaelswr@gsr:~$ sudo cat /proc/mdstat
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1] [r
aid10]
md0 : active raid6 sdg[5] sdf[4](S) sde[3] sdd[2] sdc[1]
      204800 blocks super 1.2 level 6, 512k chunk, algorithm 2 [4/4] [UUUU]

unused devices: <none>
rafaelswr@gsr:~$ █

```

5.

```

rafaelswr@gsr:~$ sudo pvs
PV          VG  Fmt Attr PSize  PFree
/dev/md0    vg0  lvm2 a-- 196,00m 32,00m
/dev/sdh    vg0  lvm2 a-- 200,00m   0
/dev/sdi    vg0  lvm2 a-- 200,00m   0
rafaelswr@gsr:~$ █

```

```

rafaelswr@gsr:~$ sudo pvdisplay -m /dev/md0
--- Physical volume ---
PV Name           /dev/md0
VG Name           vg0
PV Size           200,00 MiB / not usable 4,00 MiB
Allocatable       yes
PE Size           4,00 MiB
Total PE          49
Free PE           8
Allocated PE      41
PV UUID           UuGC58-nSLu-7ATn-0l2n-CnVu-wzPf-pM0wzJ

--- Physical Segments ---
Physical extent 0 to 0:
  Logical volume   /dev/vg0/lv1_rmeta_0
  Logical extents  0 to 0
Physical extent 1 to 13:
  Logical volume   /dev/vg0/lv1_rimage_0
  Logical extents  0 to 12
Physical extent 14 to 37:
  Logical volume   /dev/vg0/lv2
  Logical extents  0 to 47
Physical extent 38 to 40:
  Logical volume   /dev/vg0/lv0
  Logical extents  62 to 64
Physical extent 41 to 48:
  FREE

```

```
rafaelswr@gsr:~$ █
```

```

rafaelswr@gsr:~$ sudo pvdisplay -m /dev/sdh
--- Physical volume ---
PV Name           /dev/sdh
VG Name           vg0
PV Size           204,00 MiB / not usable 4,00 MiB
Allocatable       yes (but full)
PE Size           4,00 MiB
Total PE          50
Free PE           0
Allocated PE      50
PV UUID           zXezkf-MIe2-SUaX-pNjW-8CYK-hcVG-L7fXom

--- Physical Segments ---
Physical extent 0 to 49:
  Logical volume   /dev/vg0/lv0
  Logical extents  0 to 49

```

```
rafaelswr@gsr:~$ █
```

```

rafaelswr@gsr:~$ sudo pvdisplay -m /dev/sdi
--- Physical volume ---
PV Name           /dev/sdi
VG Name           vg0
PV Size           204,00 MiB / not usable 4,00 MiB
Allocatable       yes (but full)
PE Size           4,00 MiB
Total PE          50
Free PE           0
Allocated PE      50
PV UUID           Ytjpj0-xbRU-r4v3-nX6Y-rNa0-a6WK-C7DMqq

--- Physical Segments ---
Physical extent 0 to 0:
  Logical volume   /dev/vg0/lv1_rmeta_1
  Logical extents  0 to 0
Physical extent 1 to 13:
  Logical volume   /dev/vg0/lv1_rimage_1
  Logical extents  0 to 12
Physical extent 14 to 37:
  Logical volume   /dev/vg0/lv2
  Logical extents  0 to 47
Physical extent 38 to 49:
  Logical volume   /dev/vg0/lv0
  Logical extents  50 to 61

rafaelswr@gsr:~$ █

```

6.

```

rafaelswr@gsr:~$ lsblk -o NAME,SIZE
NAME    SIZE
sda      1G
sdb      45G
├─sdb1   8,7G
├─sdb2    1K
├─sdb5   3,2G
├─sdb6   976M
├─sdb7   632M
└─sdb8  31,5G
sdc      1G
sdd      1G
sde      1G
sdf      1G
sr0     1024M
rafaelswr@gsr:~$ █

```

a)

-> Habilitar para *contrib non-free*:

```
$sudo nano /etc/apt/sources.list
```

```
rafaelswr@gsr: ~  
GNU nano 5.4 /etc/apt/sources.list *  
#deb cdrom:[Debian GNU/Linux 11.5.0 _Bullseye_ - Official amd64 NETINST 20220910-10:38]/ bullseye  
deb http://deb.debian.org/debian/ bullseye main contrib non-free  
deb-src http://deb.debian.org/debian/ bullseye main contrib non-free  
  
deb http://security.debian.org/debian-security bullseye-security main contrib non-free  
deb-src http://security.debian.org/debian-security bullseye-security main contrib non-free  
  
# bullseye-updates, to get updates before a point release is made;  
# see https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#updates_and_backports  
deb http://deb.debian.org/debian/ bullseye-updates main contrib non-free  
deb-src http://deb.debian.org/debian/ bullseye-updates main contrib non-free  
  
# This system was installed using small removable media  
# (e.g. netinst, live or single CD). The matching "deb cdrom"  
# entries were disabled at the end of the installation process.  
# For information about how to configure apt package sources,  
# see the sources.list(5) manual.
```

-> Instalação dos packages:

```
$sudo apt install linux-headers-amd64
```

```
$sudo apt install zfsutils-linux
```

b)

-> Criar a *pool*, raidz1 pool (≥ 3 discos, 1 disco redundante)

```
rafaelswr@gsr: ~  
rafaelswr@gsr:~$ sudo zfs --version  
zfs-2.0.3-9  
zfs-kmod-2.0.3-9  
rafaelswr@gsr:~$ sudo zpool create userpool raidz sda sdc sdd  
rafaelswr@gsr:~$ sudo zpool add userpool log mirror sde sdf  
rafaelswr@gsr:~$ sudo zpool list  
NAME      SIZE  ALLOC  FREE  CKPOINT  EXPANDSZ  FRAG    CAP  DEDUP  HEALTH  ALTROOT  
userpool  2.75G  343K  2.75G      -          -       0%   0%   1.00x  ONLINE  -  
rafaelswr@gsr:~$ sudo zpool status  
pool: userpool  
state: ONLINE  
config:  
    NAME      STATE      READ WRITE CKSUM  
    userpool  ONLINE    0    0    0  
      raidz1-0  ONLINE    0    0    0  
        sda    ONLINE    0    0    0  
        sdc    ONLINE    0    0    0  
        sdd    ONLINE    0    0    0  
    logs  
      mirror-1  ONLINE    0    0    0  
        sde    ONLINE    0    0    0  
        sdf    ONLINE    0    0    0  
errors: No known data errors  
rafaelswr@gsr:~$
```

c)


```

rafaelswr@gsr:~$ sudo zfs list
NAME      USED  AVAIL    REFER  MOUNTPOINT
userpool  149K  1.71G    30.6K  /userpool
rafaelswr@gsr:~$ sudo zfs create userpool/users
rafaelswr@gsr:~$ sudo zfs list
NAME      USED  AVAIL    REFER  MOUNTPOINT
userpool    179K  1.71G    30.6K  /userpool
userpool/users  30.6K  1.71G    30.6K  /userpool/users
rafaelswr@gsr:~$ sudo zfs get compression userpool/users
NAME      PROPERTY  VALUE      SOURCE
userpool/users  compression  off        default
rafaelswr@gsr:~$ sudo zfs set compression=lz4 userpool/users
rafaelswr@gsr:~$ sudo zfs get compression userpool/users
NAME      PROPERTY  VALUE      SOURCE
userpool/users  compression  lz4        local
rafaelswr@gsr:~$ █

```

d)

```

rafaelswr@gsr:~$ sudo zfs list
NAME      USED  AVAIL    REFER  MOUNTPOINT
userpool    181K  1.71G    30.6K  /userpool
userpool/users  30.6K  1.71G    30.6K  /userpool/users
rafaelswr@gsr:~$ █

```

e)

IO benchmark (questão 6):

Velocidade de escrita em torno do 800MB/s e de leitura nos 1,3GB/s.

```

rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$ escrita
rafaelswr@gsr:~$ sudo dd if=/dev/zero of=/userpool/users/test bs=100M count=1 oflag=direct
1+0 registros dentro
1+0 registros fora
104857600 bytes (105 MB, 100 MiB) copiados, 0,119541 s, 877 MB/s
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$ leitura
rafaelswr@gsr:~$ sudo dd if=/userpool/users/test of=/dev/null bs=100M count=1 iflag=direct
1+0 registros dentro
1+0 registros fora
104857600 bytes (105 MB, 100 MiB) copiados, 0,0817818 s, 1,3 GB/s
rafaelswr@gsr:~$ █

```

IO benchmark (questão 4):

Velocidade de escrita em torno do 400MB/s e de leitura nos 1,1GB/s.

```

rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$ sudo dd if=/dev/zero of=/mnt/vg0-lv0/test bs=100M count=1 oflag=direct
1+0 registros dentro
1+0 registros fora
104857600 bytes (105 MB, 100 MiB) copiados, 0,258538 s, 406 MB/s
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$ sudo dd if=/mnt/vg0-lv0/test of=/dev/null bs=100M count=1 iflag=direct
1+0 registros dentro
1+0 registros fora
104857600 bytes (105 MB, 100 MiB) copiados, 0,0985863 s, 1,1 GB/s
rafaelswr@gsr:~$
rafaelswr@gsr:~$
rafaelswr@gsr:~$ █

```

escrita

leitura

7.

Deduplication tem como objetivo eliminar cópias duplicadas de dados repetidos ou redundantes. Esta técnica é usada para melhorar a utilização do armazenamento, garantindo que apenas existe uma instância de dados em armazenamento e substituindo blocos de dados redundantes por apontadores para a cópia de dados exclusiva.

8.

O Time Machine é uma aplicação disponível para Apple MacOS que permite efetuar cópias de segurança de todos os ficheiros para um dispositivo de armazenamento externo para que posteriormente possam ser restaurados. Três principais características:

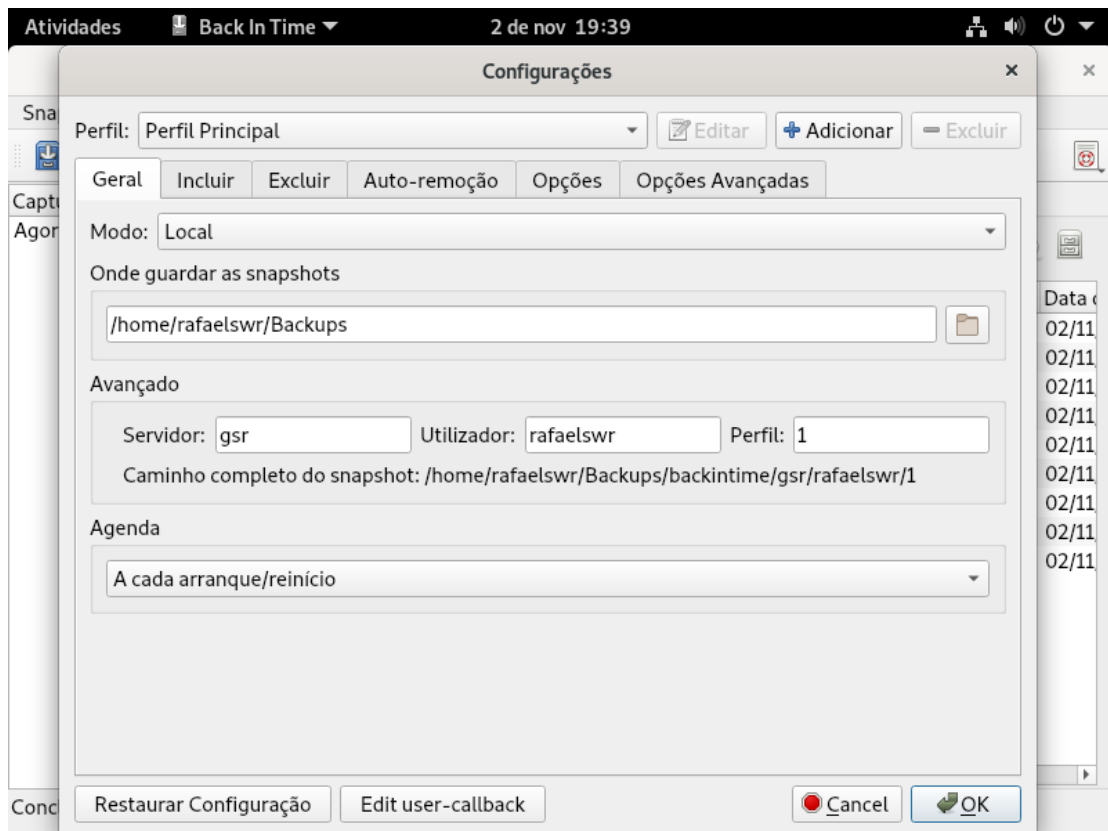
- Capacidade de fazer backups contínuos de forma automática, mantendo o histórico de backups;
- Fazer backups na unidade de rede através do AirPort;
- Fazer backups incrementais da máquina usando uma drive externa;

9.

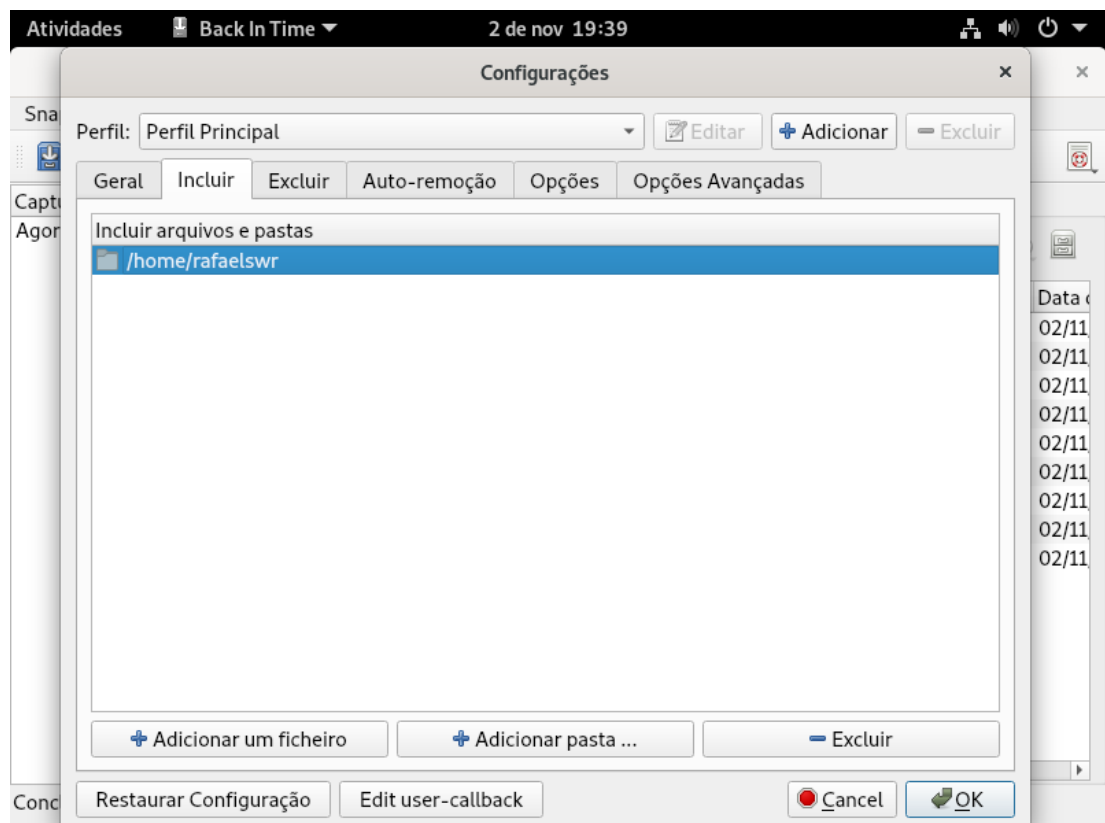
Depois de instalar a Debian VM, na linha de comandos instalamos o backintime-qt (solução de backup de sistema para sistemas linux) com o comando abaixo:

```
$ sudo apt-get install backintime-qt
```

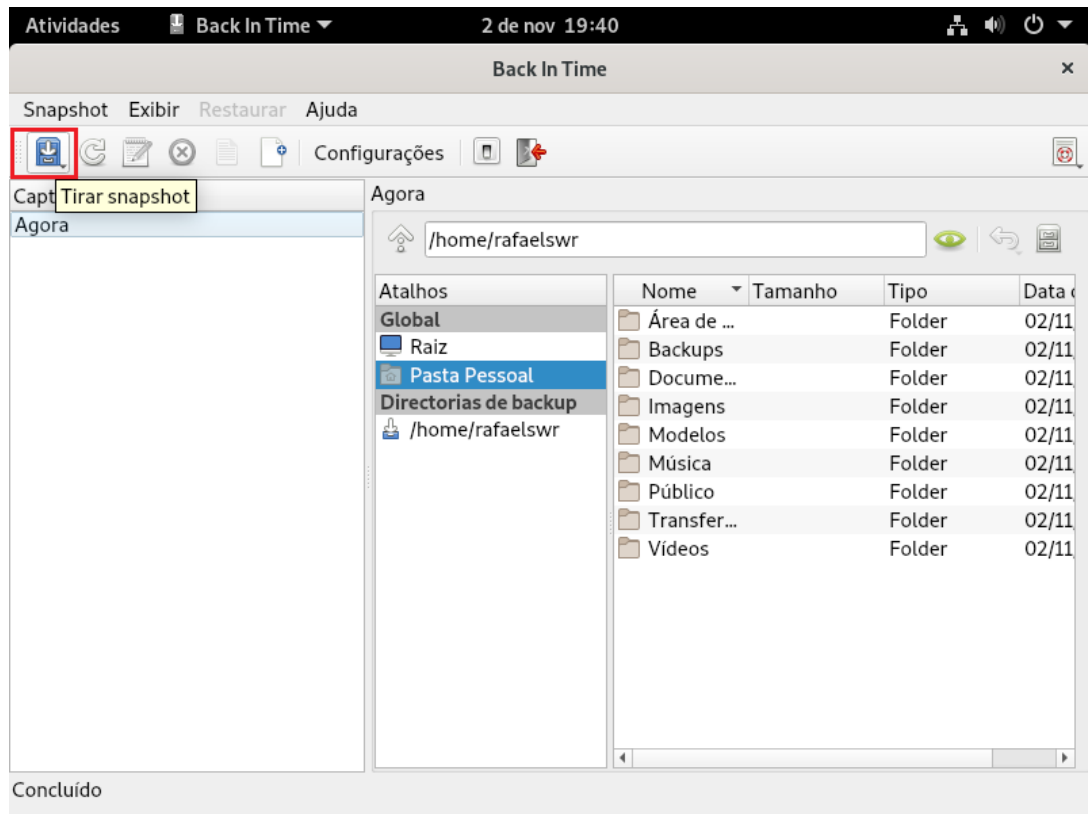
- Ao iniciar aparece a janela de configurações do Back In Time, onde podemos criar perfis diferentes com configurações de backup separadas, escolher onde guardar os nossos snapshots e agendar os nossos backups.



- No perfil selecionado, podemos incluir os arquivos e pastas que queremos guardar no nosso backup.



- Depois das configurações de perfil, podemos tirar um simples snapshot carregando no ícone.



- Depois de concluído o nosso snapshot á diretoria de backup, a referência com a data e hora é mostrada.

